

STATEMENT OF WORK

AI Buddy — AI-Powered Student Study Assistant

Revised Delivery Timeline & Resource Assignment

Field	Detail
Client / Owner	Caper Pvt LTD
Project	AI Buddy — AI-Powered Student Study Assistant
SoW Version	v1.0
Project Start Date	Monday, 31 August 2026
Project End Date (Hard Deadline)	Tuesday, 29 September 2026
Total Duration	30 calendar days — 3 sprints of 10 days each
Status	Draft — Pending Client Sign-off

1. Purpose & Background

This Statement of Work (SoW) defines the revised delivery timeline, sprint structure, and resource assignment for the AI Buddy platform, superseding the open-ended schedule implied in the original Software Requirements Document (SRD v2.0). The project scope — 28 application screens, 13 MongoDB collections, a Gemini/Groq-based NotebookLLM pipeline, an 8-trigger email system, exam lockout, ethics detection, and a student safety module — remains as specified in the SRD. This SoW compresses delivery into a fixed 30-day, 3-sprint schedule ending 29 September 2026, and assigns every workstream to a named resource.

This document is the binding reference for sprint planning, resource commitment, and acceptance at each milestone. Any change to scope, timeline, or resourcing must be raised as a formal Change Request (CR) per Section 7.

2. Project Team & Coding Responsibilities

All four team members hold delivery (coding) responsibility in addition to their process role. Coding ownership is assigned by module so the same person carries a feature from build through hardening across sprints, minimizing handoff risk.

Name	Role	Primary Coding Ownership	Process Responsibility
Hitarth Patel	Solution Architect & Tech Lead	Auth/RBAC, LLM & RAG core, ethics classifier, performance	Architecture decisions; required reviewer & merger on every pull request
Om Pandey	Scrum Master	Data schemas, mock test & learning path services, exam lockout	Sprint planning, standups, blocker log, QA sign-off before each sprint close
Shivam Mishra	Project Manager	Mailer system, faculty booking, admin dashboard	Client communication, Change Request intake & triage, UAT coordination
Rehan Mulani	Business Analyst	Content ingestion pipeline, forum & feedback, safety module	Requirement clarification, RAG content tagging, acceptance criteria

Source-control convention:

- Every individual commits their own code on their own feature branch (feature/sprintN-<module>-<initials>). No team member commits work on behalf of another.
- Hitarth Patel is the required reviewer and sole merger of pull requests into the develop branch.
- Om Pandey signs off on QA before develop is merged into main at the end of each sprint.

3. Timeline Overview

Sprint	Dates	Duration	Buffer (CR / bug-fix)	Theme
Sprint 1	31 Aug – 09 Sep 2026	10 days	08–09 Sep (2 days)	Foundation & Core Infrastructure
Sprint 2	10 Sep – 19 Sep 2026	10 days	18–19 Sep (2 days)	Core Feature Build
Sprint 3	20 Sep – 29 Sep 2026	10 days	26–29 Sep (4 days)	Safety, Compliance, Integration & Hardening

Buffer days are reserved capacity for Change Requests and defect resolution — they are not available for pulling forward new feature work. Sprint 3's buffer is intentionally widest (4 days) because it directly precedes the hard deadline and production release.

4. Sprint 1 — Foundation & Core Infrastructure

31 August – 09 September 2026

Objective: establish the data, auth, and integration layers that every later feature depends on. Nothing here needs to be end-user polished, but nothing here can slip without delaying Sprints 2 and 3.

Owner	Deliverables	Upstream Dependency	Unblocks
Hitarth Patel	JWT auth, RBAC middleware, dept-scope middleware, Gemini + Groq client config with automatic fallback	None	All protected routes (Sprints 2–3)
Om Pandey	All 13 Mongoose schemas, MongoDB Atlas connection with retry logic, environment variable validation	None	Every feature (Sprints 2–3)
Shivam Mishra	SMTP transporter + generic mail-send service, application shell (Navbar/Sidebar/Footer), Login & Unauthorized pages, SSO redirect flow	Hitarth's auth layer	Booking emails (Sprint 2), all later email triggers (Sprint 3)
Rehan Mulani	Content ingestion pipeline (parse → chunk → tag → embed → upsert to Atlas Vector Search), Question Bank seed data, shared Zod validation schemas	Om's schemas	NotebookLLM / MCQ & Practice Sheet generation (Sprint 2)

Sprint 1 Exit Criteria:

- A student can authenticate via SSO and land on a role-correct dashboard shell.
- All 13 collections exist and are reachable in MongoDB Atlas.
- The ingestion pipeline runs end-to-end on at least one sample document.
- A test email sends successfully through the institution's SMTP server.

5. Sprint 2 — Core Feature Build

10 September – 19 September 2026

Objective: deliver every screen a student, faculty member, or HOD uses day-to-day, with basic (not yet fully escalated) versions of the ethics and distress checks running from day one — so these guardrails are never bolted on retroactively.

Owner	Deliverables	Upstream Dependency	Unblocks
Hitarth Patel	NotebookLLM service (embed → vector search → context assembly), MCQ & Practice Sheet generation service, AI Tutor chat endpoint, basic ethics keyword-check on every LLM call (block & log only)	Sprint 1: auth, LLM config, ingestion	Sprint 3 ethics escalation, hallucination reporting
Om Pandey	Mock Test generate/submit/scoring service, Learning Path generation service, Student Dashboard, Question Bank browse page	Sprint 1: schemas	Mock Test Results page; Sprint 3 lockout logic
Shivam Mishra	Faculty Availability CRUD, Doubt Session book/confirm/decline flow, meeting-link service, E-01/E-02/E-03 emails wired in, Faculty pages (4), Admin Timetable upload page (CSV parse, no enforcement yet)	Sprint 1: mailer config	Sprint 3 lockout enforcement (uses this timetable data)
Rehan Mulani	Forum CRUD, Grievance flag + E-06 email, Weekly Feedback form + cron trigger, Admin Feedback Viewer, HOD Audit Log (read-only) + Faculty Management page, basic distress keyword-check on AI Tutor (log only)	Sprint 1: schemas, mailer; Hitarth's chat endpoint	Sprint 3 safety escalation, ethics config UI

Sprint 2 Exit Criteria:

- Students can generate MCQs, take a mock test, view results, and chat with the AI Tutor.
- Students can book a faculty session and receive a confirmation email; faculty can confirm/decline sessions.
- Forum posting and weekly feedback submission work end-to-end.
- HOD can view (read-only) audit log entries.
- Exam lockout and full safety/ethics escalation are intentionally deferred to Sprint 3.

6. Sprint 3 — Safety, Compliance, Integration & Hardening

20 September – 29 September 2026

Objective: close every loop opened in Sprint 2 — escalation emails, lockout enforcement, ethics/distress hardening — then dedicate the final four days exclusively to regression testing, client Change Requests, and production readiness. No new feature branches open after 25 September.

Owner	Deliverables	Upstream Dependency
Hitarth Patel	Secondary Gemini ethics classifier for inconclusive keyword matches, EthicsFlag → BullMQ → E-04 escalation email (5-minute SLA), NotebookLLM/MCQ & Practice Sheet performance tuning (P95 ≤ 3s), final architecture review across all merged branches	Sprint 2 basic ethics check
Om Pandey	Exam lockout cron (15-min sweep) + enforcement middleware, manual unlock endpoint with audit trail, Admin Dashboard KPI aggregates, end-to-end QA pass and regression test execution	Sprint 2 timetable data (Shivam)
Shivam Mishra	Remaining email templates (E-05, E-07, E-08) with unsubscribe footer on E-08, BullMQ retry wiring for guaranteed delivery, Admin User Management, Change Request triage for the sprint	Sprint 1 mailer, Sprint 2 templates
Rehan Mulani	Two-step distress confirmation flow + Emergency Panel (24/7, lockout-exempt) with E-07 wiring, Hallucination Report page with E-05 wiring, HOD Ethics Config page, UAT support with client	Sprint 2 basic distress check

Sprint 3 Exit Criteria (Release Gate):

- Exam lockout correctly blocks AI routes within the computed window; the Emergency Panel remains accessible during lockout.
- All 8 email triggers fire correctly, including the 5-minute ethics escalation SLA and 2-minute distress SLA.
- A full regression pass across all 28 pages and the RBAC matrix shows zero cross-role or cross-department data leakage.
- Zero open blocker-severity defects before 26 September.
- 26–29 September is reserved for Change Requests, bug-fixing, and client UAT sign-off only.

7. Change Request Handling

Each sprint's buffer window (2, 2, and 4 days respectively) is reserved capacity for Change Requests and defect resolution — it is not available for pulling forward feature work from later sprints.

- Shivam Mishra (PM) logs all incoming Change Requests as they arrive during the sprint.
- Triage and implementation of approved CRs happens only within that sprint's buffer window.
- Any CR too large to fit the remaining buffer is queued to a post-launch v1.1 backlog rather than compressed into the 29 September deadline.

8. Why This Sequencing Protects Testing & Production

- Schemas (Sprint 1, Om Pandey) are built before anything that reads or writes them.
- Auth and RBAC (Sprint 1, Hitarth Patel) are built before any protected route in Sprints 2–3.

- The mailer is split in two: basic send capability lands in Sprint 1, booking-related templates in Sprint 2, and the full 8-template suite with guaranteed-delivery retries in Sprint 3 — preventing the email system from being built too late to test properly.
- Ethics and distress checks ship in the same sprint as the AI Tutor feature they guard (Sprint 2); only the escalation/notification layer is deferred to Sprint 3, so the guardrail itself is never bolted on afterward.
- Timetable upload (Sprint 2, Shivam Mishra) and lockout enforcement (Sprint 3, Om Pandey) are deliberately split across two owners and two sprints, so lockout logic is tested against real uploaded data rather than stubs.
- The final four days of Sprint 3 (26–29 September) are protected as buffer only, with no competing feature work — this is the direct control against defects surfacing at review or in production.

9. Assumptions & Constraints

- All four team members are available full-time for the entire 30-day window.
- Institutional SSO, SMTP credentials, and academic content (syllabi/notes/past papers) for ingestion are made available to the team no later than Sprint 1, Day 1.
- Gemini and Groq free-tier API keys are provisioned before Sprint 1 begins.
- No scope additions are made outside the formal Change Request process defined in Section 7.
- Sign-off on each sprint's exit criteria is required from Om Pandey (QA) before the next sprint's feature work begins.

10. Approval & Sign-off

By signing below, both parties agree to the timeline, scope, and resource assignment described in this Statement of Work.

Name	Role	Signature	Date
Sayantan Ghosh	Client		
Shivam Mishra	Project Manager — Caper Pvt LTD		
Om Pandey	Scrum Master — Caper Pvt LTD		
Rehan Mulani	Business Analyst — Caper Pvt LTD		
Hitarth Patel	Solution Architect & TL — Caper Pvt LTD		

Document Status: ☐ Draft ☐ Under Review ☐ Approved ☐ Rejected